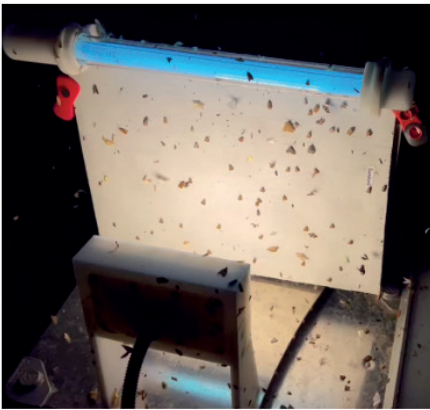


Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring

Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring

... in Automated Moth Monitoring



(1) Moth Camera Trap

Species described on earth



Insects play essential role in terrestrial ecosystem

Unknown extent of species extinction
and the effects of climate change on population

Key challenge: Monitoring of insects by
specialized experts, who are limited in number

Developments in machine learning have the potential
to develop scalable monitoring methods.

(1) Aditya, Cunha, Bunsen, Pasi, Viklund, Larrivee, Rolnick. 2023. "A Machine Learning Pipeline for Automated Insect Monitoring." in Climate Change AI

(2) Wirth, Christian, Helge Bruelheide, Nina Farwig, Jori Marx, and Josef Settele, eds. 2024. Faktencheck Artenvielfalt: Bestandsaufnahme und Perspektiven für den Erhalt der biologischen Vielfalt in Deutschland. München: oekom verlag.

Challenges ... in Automated Moth Monitoring



Fine-Grained Species Classification

More than 1,100 species of large and even more than 4,000 smaller species of nocturnal moths



High Res.



Low Res.

Domain Generalisation

High-resolution training data and low-resolution test data from moth traps



unknown species

Novelty Detection

Identification of new unknown species due to climate change or invasive species

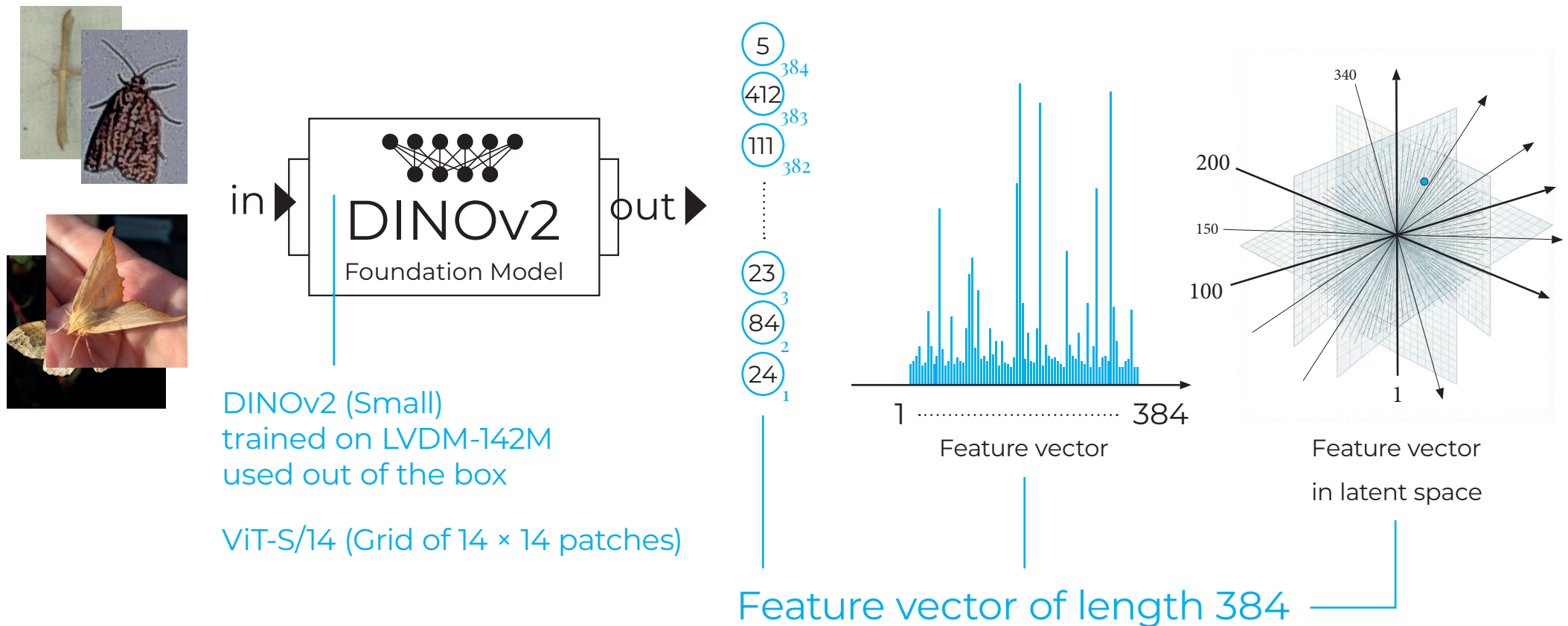


Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring

Adapting DINOv2 Embeddings

Foundation Model encodes images into feature vectors

Embedding represents the Image

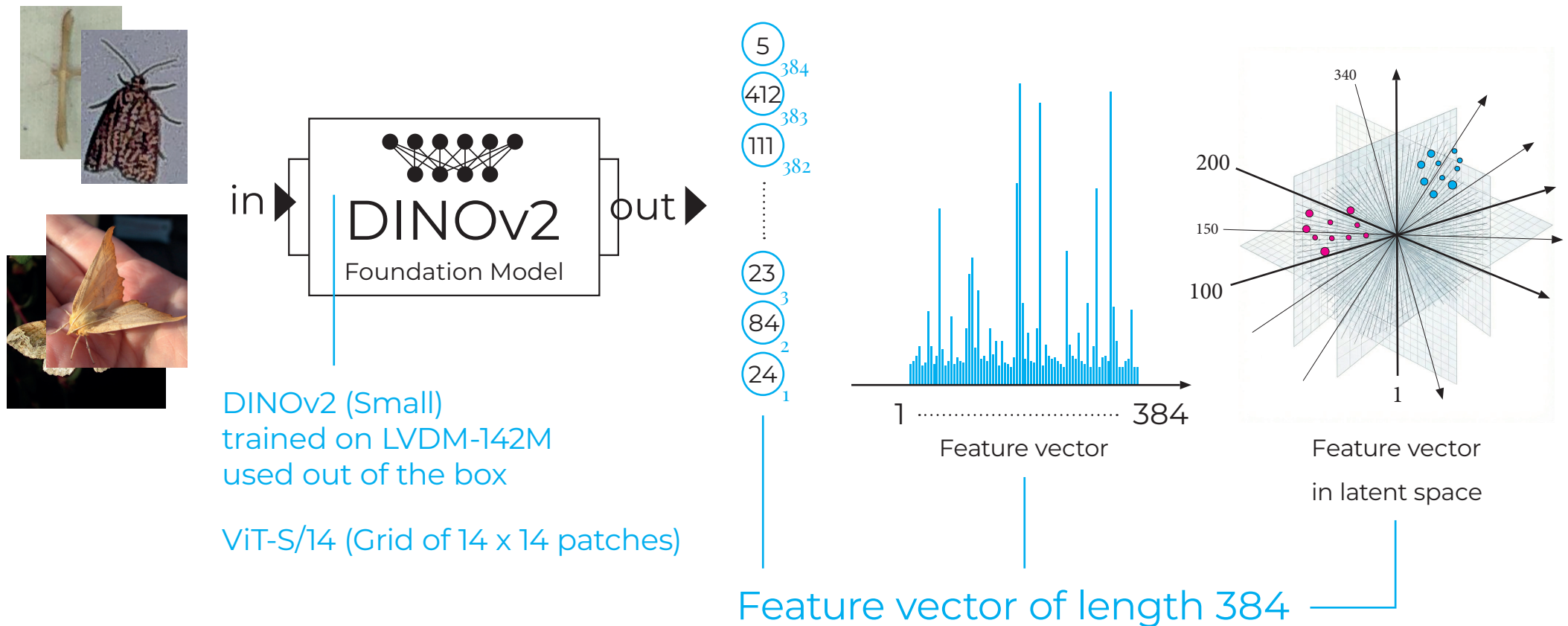


- (4) Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. 2024. "DINOv2: Learning Robust Visual Features without Supervision." 32. doi:10.48550/arXiv.2304.07193.

Adapting DINOv2 Embeddings

Foundation Model encode images into feature vectors

Embedding represents the Image



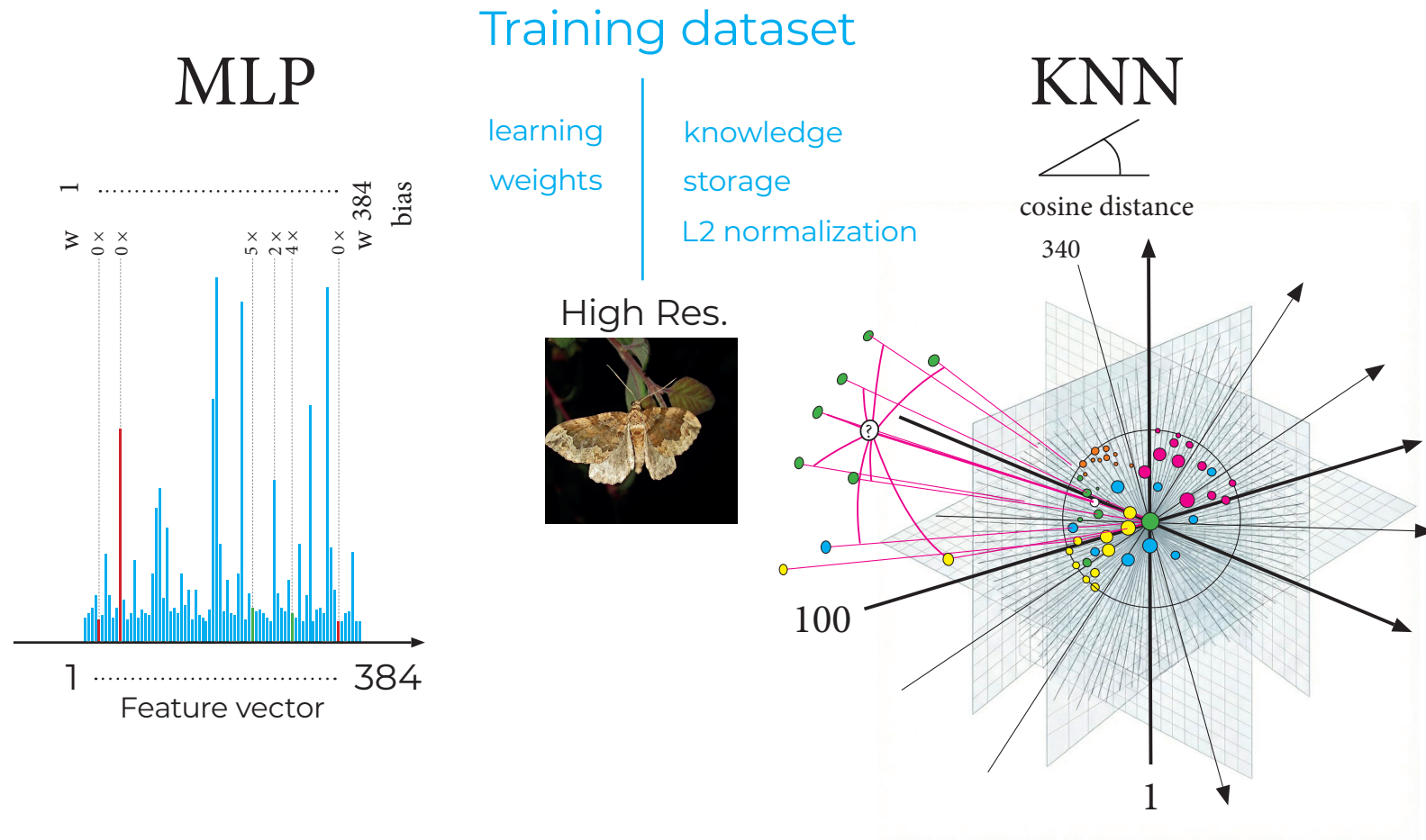
- (4) Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. 2024. "DINOv2: Learning Robust Visual Features without Supervision." 32. doi:10.48550/arXiv.2304.07193.



Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring



& for 91 Fine-Grained Species Classification Domain Generalisation



& for 91 Fine-Grained Species Classification Domain Generalisation

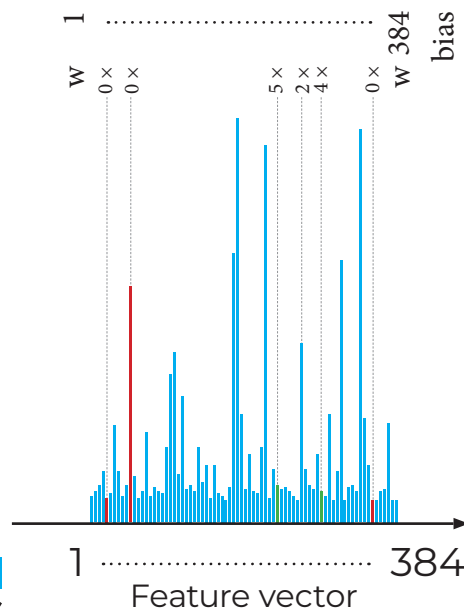
Test dataset

High Res.



94.33%

MLP



Training dataset

learning
weights

knowledge
storage

L2 normalization

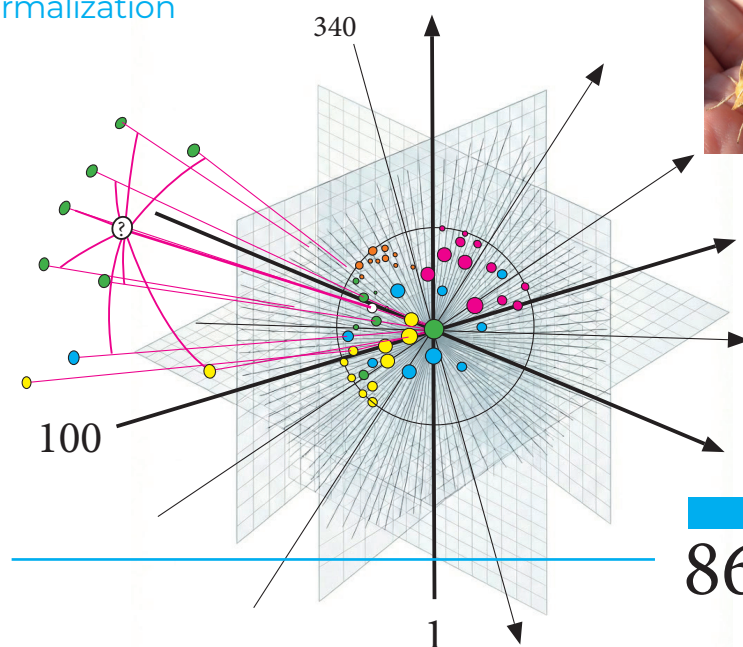
High Res.



(-8.12)

KNN

cosine distance



Test dataset

High Res.



86.21%

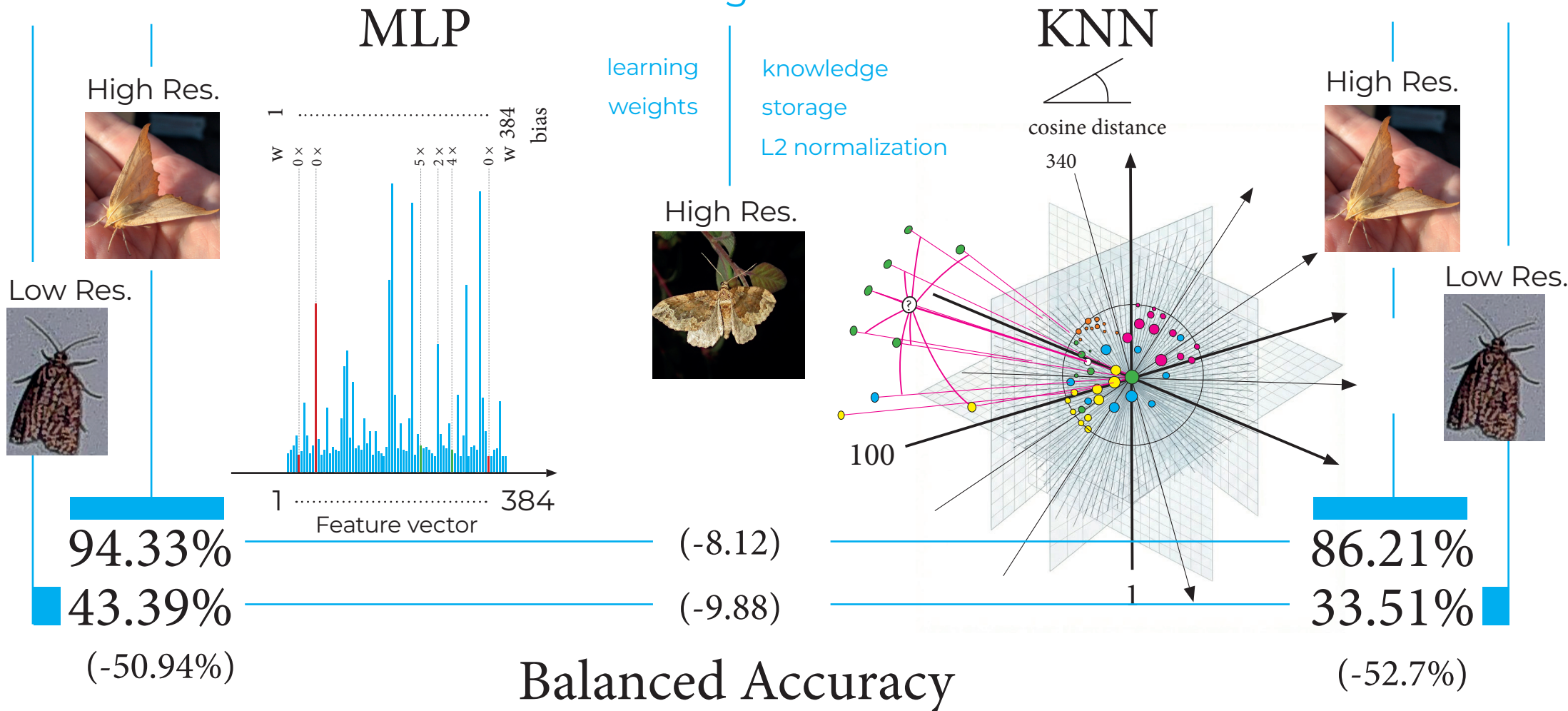
Balanced Accuracy

& for 91 Fine-Grained Species Classification Domain Generalisation

Test dataset

Training dataset

Test dataset

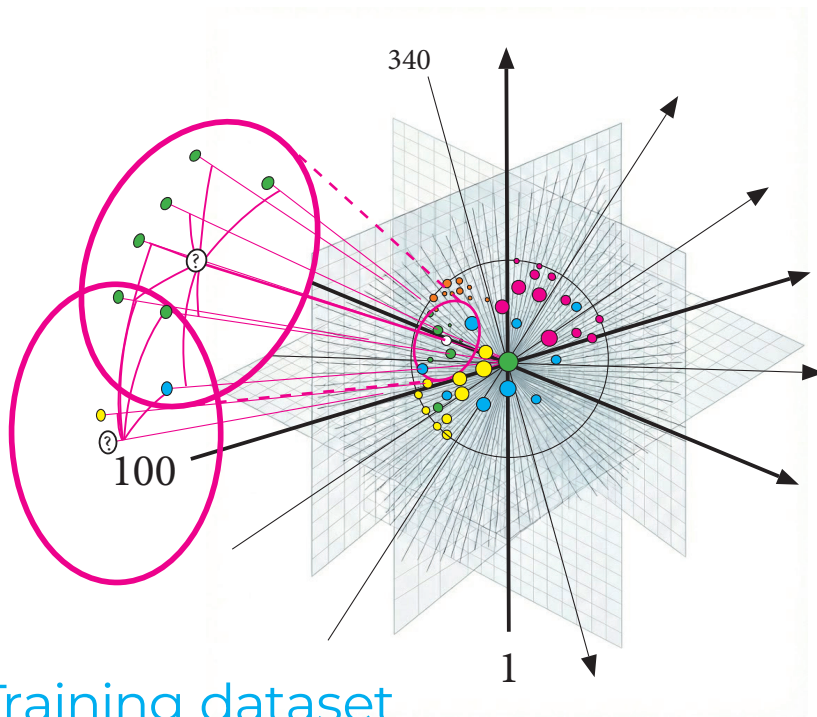


Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring



Deep Nearest Neighbor – (91 ID vs. 37 OOD)

Threshold based differentiation between
ID and OOD Samples



Training dataset

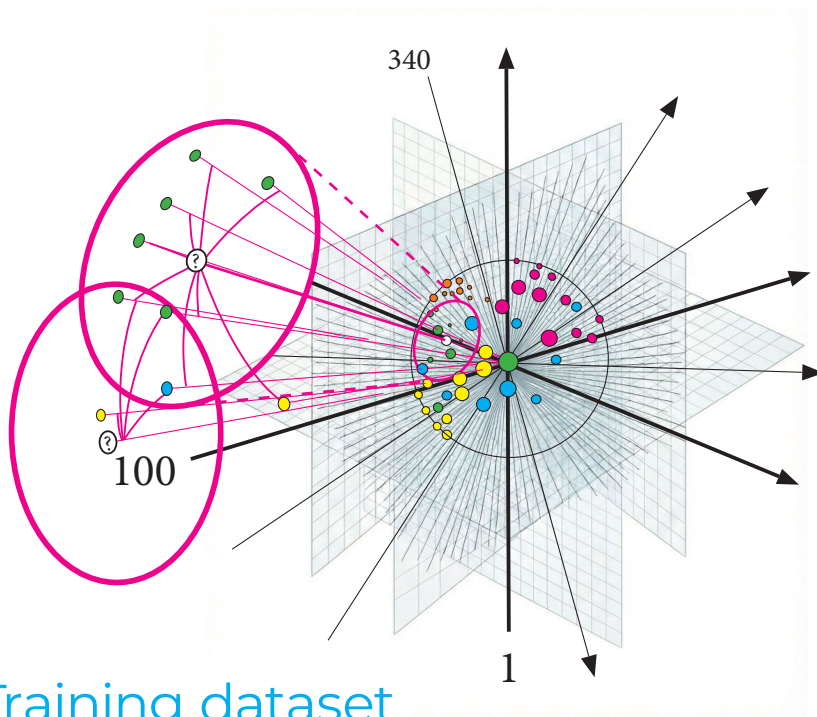
knowledge storage

L2 normalization

Deep Nearest Neighbors – (91 ID vs. 37 OOD)

defining thresholds

Threshold based differentiation between ID and OOD Samples



Training dataset

knowledge storage

L2 normalization

■ over_all_examples based on validation dataset

- every correct predicted Sample in Validation dataset
- save distance to k-th neighbour in a list
- sort distances and take value by percentile

list = [0.15, 0.17, 0.23, 0.26, 0.29, 0.32, 0.52, 0.63, 0.66, 0.73]



■ for_each_species based on validation dataset

- save distance to k-th neighbour for each class in a list
- sort distances and take value by percentile for each class

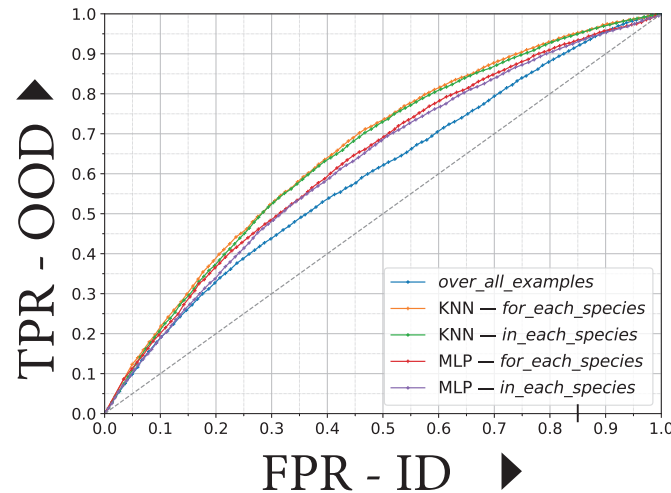
Deep Nearest Neighbors – (91 ID vs. 37 OOD)

Test dataset

High Res.



ROC-CURVE



AUROC

over_all_examples 0.5963 (-0.0562)

combined with KNN

for_each_species 0.6523

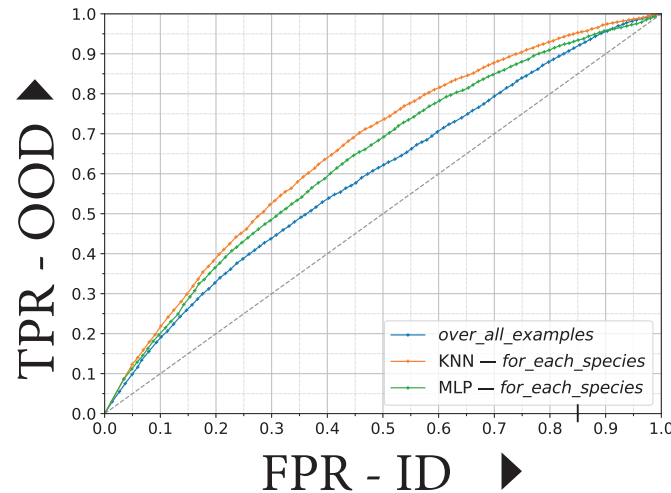
Deep Nearest Neighbors – (91 ID vs. 37 OOD)

Test dataset

High Res.



ROC-CURVE



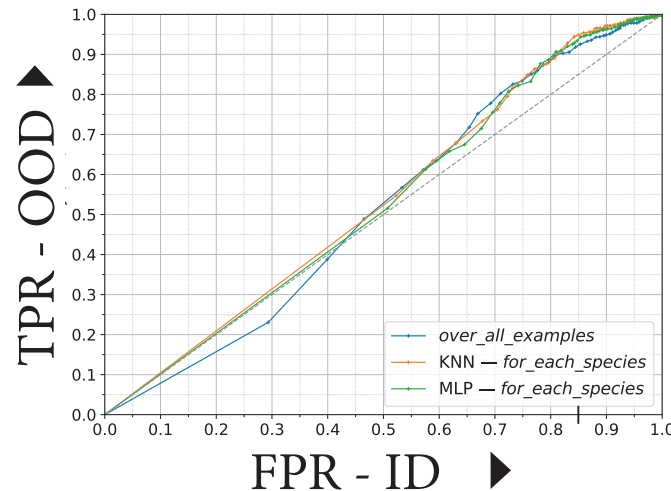
AUROC

over_all_examples 0.5963 (-0.0560)

combined with KNN

for_each_species 0.6523

Low Res.

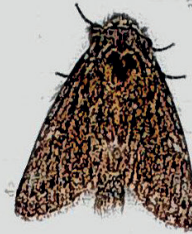


AUROC

over_all_examples 0.5169 (-0.1354)

combined with KNN

for_each_species 0.5350 (-0.1173)

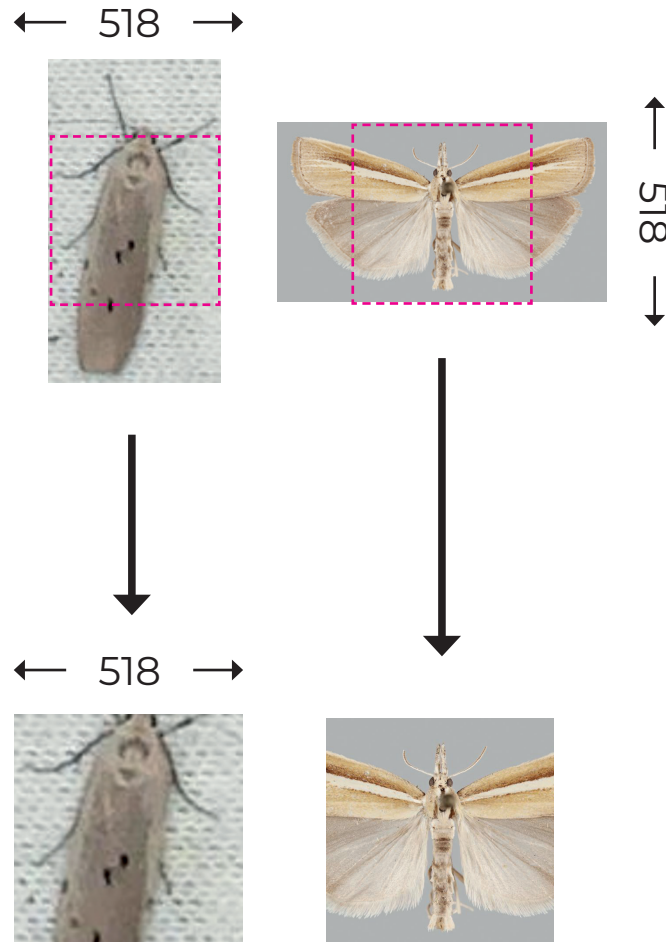


Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring



Adapting DINOv2 Embeddings

preprocessing pipeline



Rectangular images are cropped:

shorter side is scaled to 518 pixel

followed by 518×518 pixel center crop

Loss of Information:

about the shape of the Moth

less information encoded in embeddings

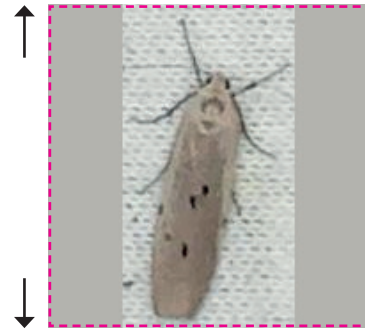
indicators for differentiation of species are lost

Adapting Images to Avoid Information Loss

Original Image



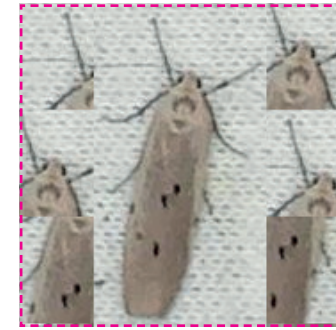
Mean Color



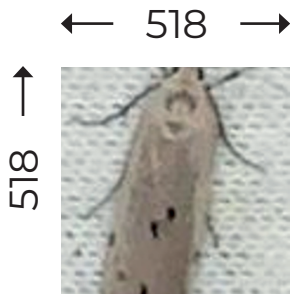
Small Blur



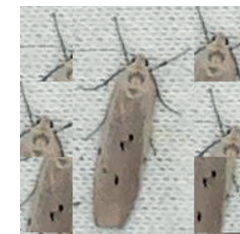
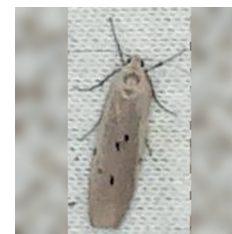
Large Patches



size in px



Cropped images after preprocessing pipeline



Adapting Images to Avoid Information Loss

Original Image



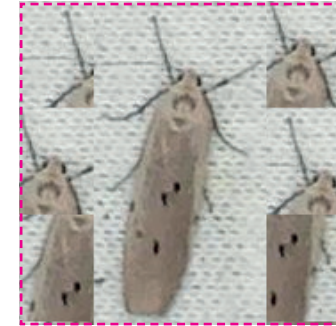
Mean Color



Small Blur



Large Patches



size in px

← 518 →

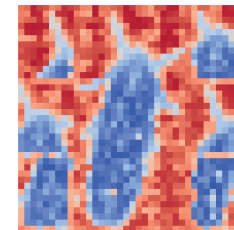
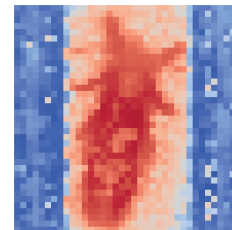
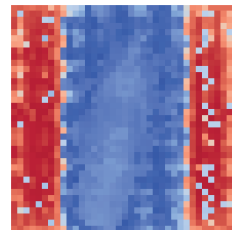
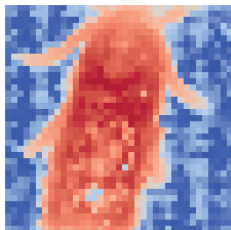
↑ 518 ↓



Cropped images after preprocessing pipeline



Visualize first PCA after preprocessing pipeline



Adapting DINOv2 Embeddings
for Fine-Grained Species Classification
and Novelty Detection
in Automated Moth Monitoring

& for Fine-Grained Species Classification Domain Generalisation

MLP

Training dataset

KNN

Test dataset
Squared

Test dataset
Squared

High Res.

High Res.

Low Res.

Low Res.



94.33% (-2.01%)
43.39% (+0.39%)

92.32% (-10.65%)
43.78% (-8.24%)

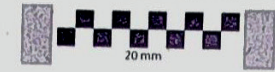
81.67% (-4.54%)
35.54% (+2.03%)

86.21%
33.51%



(5) Doerrich, Sebastian, Tobias Archut, Francesco Di Salvo, and Christian Ledig. 2024. "Integrating kNN with Foundation Models for Adaptable and Privacy-Aware Image Classification." Pp. 1–5 in 2024 IEEE International Symposium on Biomedical Imaging (ISBI).

Adapting DINOv2 Embeddings for Fine-Grained Species Classification and Novelty Detection in Automated Moth Monitoring



and Novelty Detection

with DKNN

Test dataset

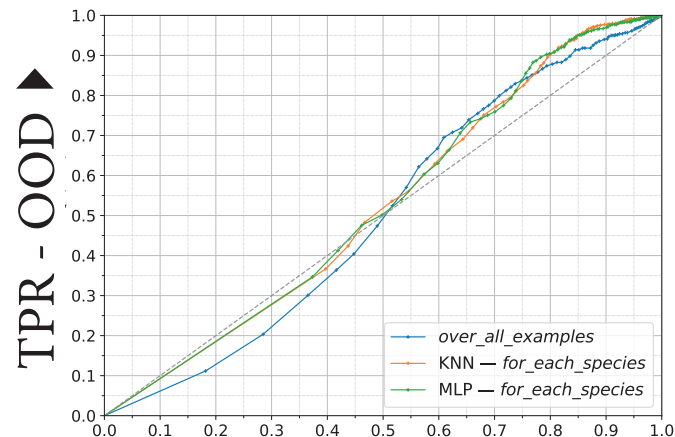
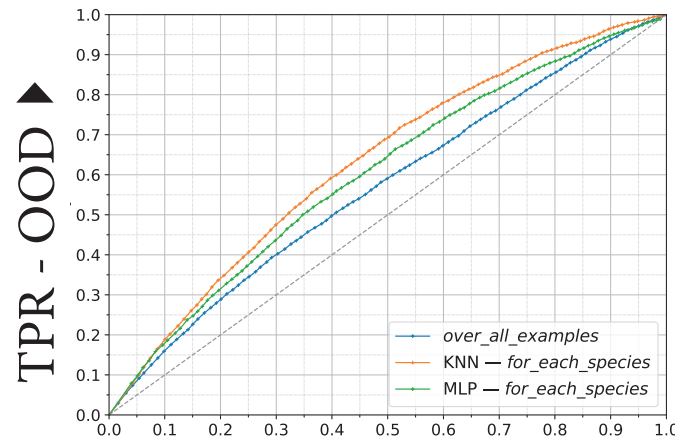
High Res.



Low Res.



ROC-CURVE



TPR - OOD ►

AUROC

■ over_all_examples 0.5681 (-0,0280)

combined with KNN

■ for_each_species 0.6209 (-0.0314)

■ over_all_examples 0.5025 (-0,0142)

combined with MLP

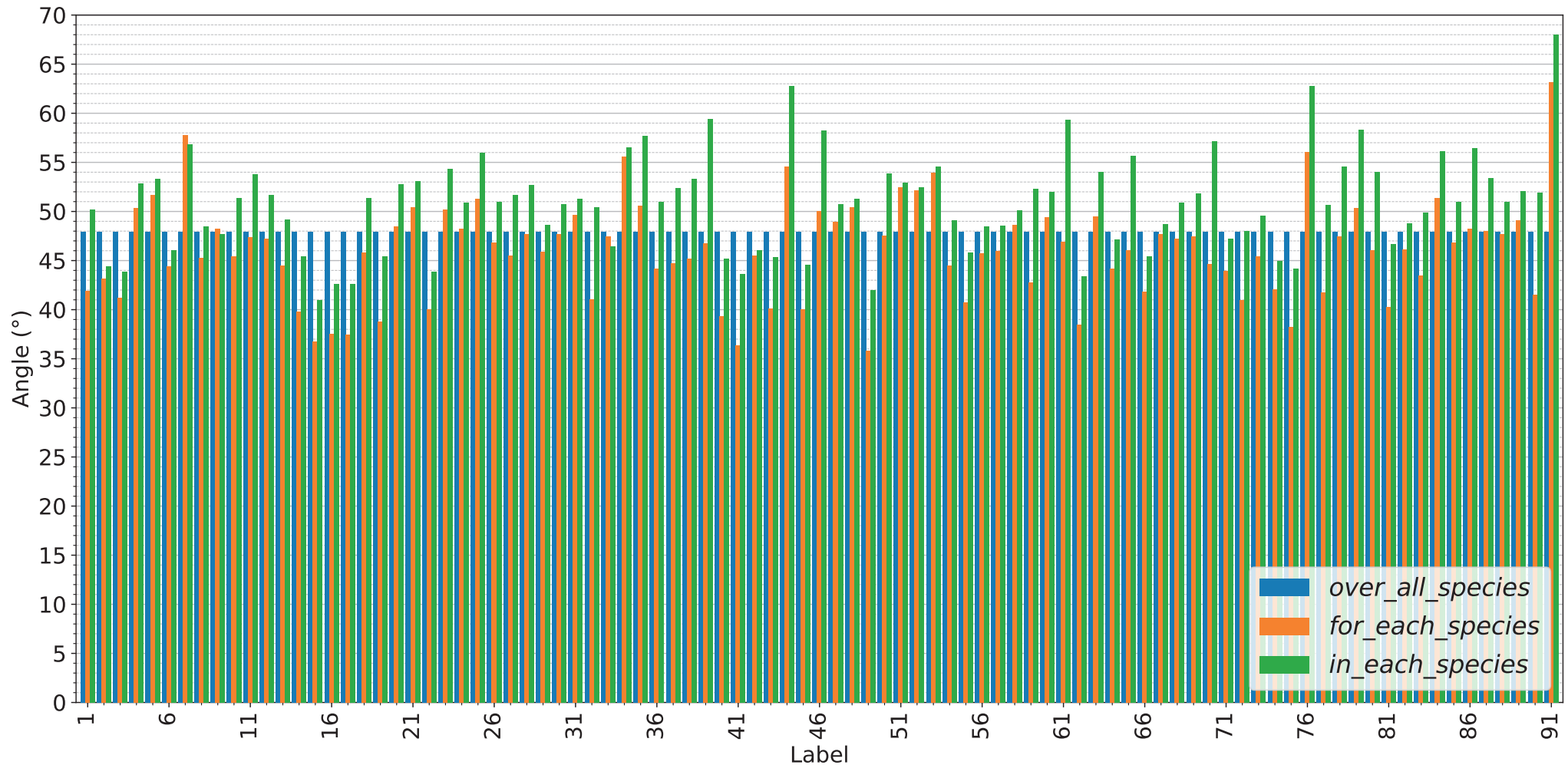
■ for_each_species 0.5243 (-0,0037)

Sources

- (1) Aditya, Cunha, Bunsen, Pasi, Viklund, Larrivee, Rolnick. 2023. “A Machine Learning Pipeline for Automated Insect Monitoring.” in Climate Change AI
- (2) Wirth, Christian, Helge Bruelheide, Nina Farwig, Jori Marx, and Josef Settele, eds. 2024. Faktencheck Artenvielfalt: Bestandsaufnahme und Perspektiven für den Erhalt der biologischen Vielfalt in Deutschland. München: oekom verlag.
- (3) Aditya, Cunha, Bunsen, Cañas, Pasi, Pinoy, Helsing, Russo, Botham, Sabourin, Fréchette, Ancil, Lopez, Navarro, Perez Pimentel, Zamora, Silva, Gagnon, August, Bjerre, Segura, Bélisle, Basset, McFarland, Roy, Høye, Larrivée, Rolnick. 2024. “Insect Identification in the Wild: The AMI Dataset.”
- (4) Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. 2024. “DINOv2: Learning Robust Visual Features without Supervision.” 32. doi:10.48550/arXiv.2304.07193.
- (5) Doerrich, Sebastian, Tobias Archut, Francesco Di Salvo, and Christian Ledig. 2024. “Integrating kNN with Foundation Models for Adaptable and Privacy-Aware Image Classification.” Pp. 1–5 in 2024 IEEE International Symposium on Biomedical Imaging (ISBI).
- (6) Sun, Yiyu, Yifei Ming, Xiaojin Zhu, and Yixuan Li. 2022. “Out-of-Distribution Detection with Deep Nearest Neighbors.” Pp. 20827–40 in Proceedings of the 39th International Conference on Machine Learning. PMLR.
- (7) Leibniz-Institut zur Analyse des Wandels der Biodiversität (LIB). n.d. “LEPMON – Erfassung Der Biodiversität von Nachtfaltern (Lepidoptera) Mit Automatisierten Kamerafallen Und Künstlicher Intelligenz.” Retrieved July 7, 2025. <https://lepmon.de/>.

and Novelty Detection (ID vs OOD)

Three distinct thresholds



Generated Datasets

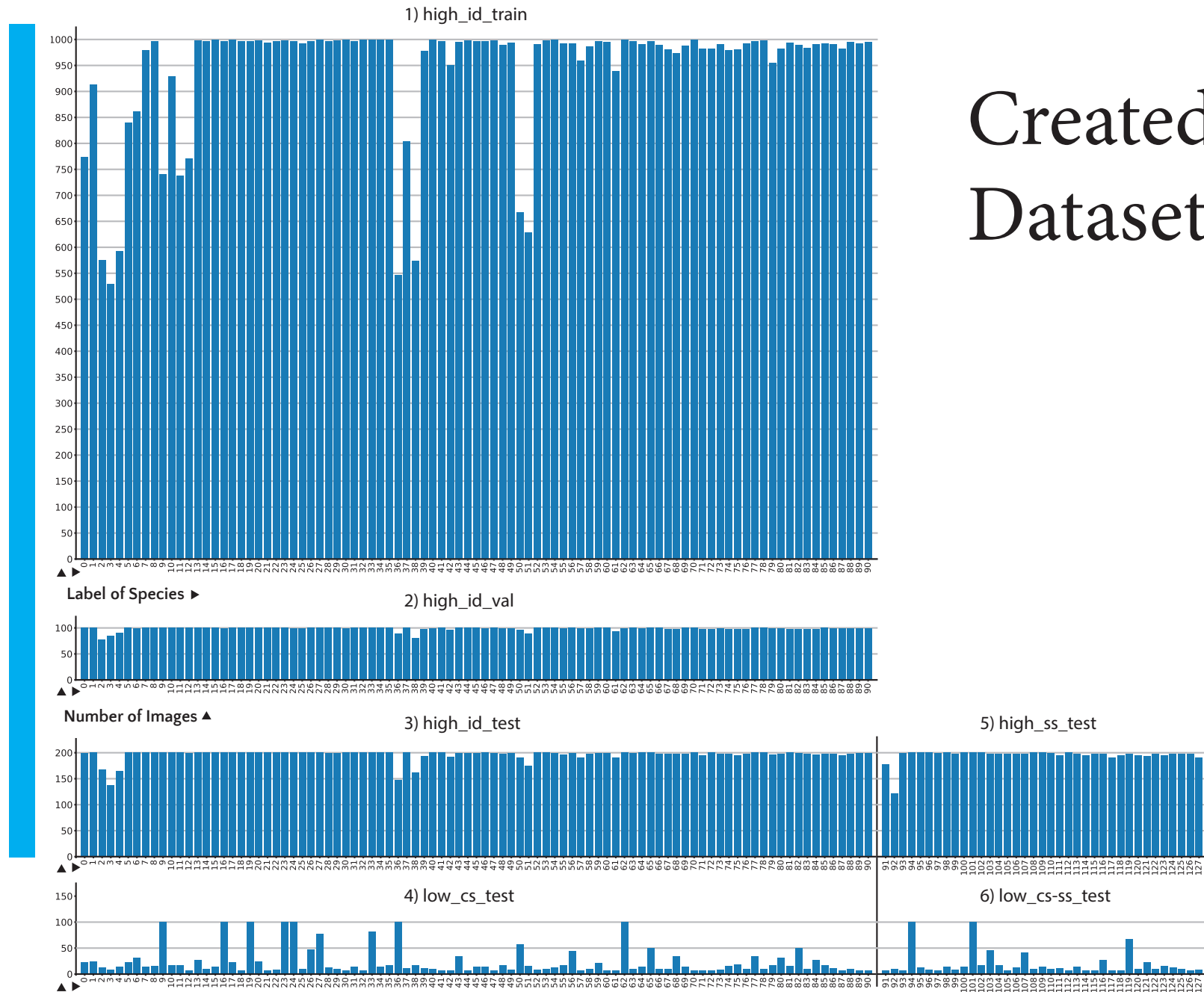
Dataset Type	Resolution	Filename	Classes	min	max	total
Training	high (GBIF)	high_id_train.csv	91	529	1000	85,803
Validation	high (GBIF)	high_id_val.csv	91	78	100	8,949
Test FG	high (GBIF)	high_id_test.csv	91	138	200	17,832
Test FG + DG	low (Trap)	low_id_test.csv	91	6	100	2,138
Test ND	high (GBIF)	high_ood_test.csv	37	122	200	7,213
Test ND + DG	low (Trap)	low_ood_test.csv	37	6	100	698

(2) Aditya, Cunha, Bunsen, Cañas, Pasi, Pinoy, Helsing, Russo, Botham, Sabourin, Fréchette, Anctil, Lopez, Navarro, Perez Pimentel, Zamora, Silva, Gagnon, August, Bjerger, Segura, Bélisle, Basset, McFarland, Roy, Høye, Larrivée, Rolnick. 2024. “Insect Identification in the Wild: The AMI Dataset.”



High Res.

Created Datasets



Low Res.

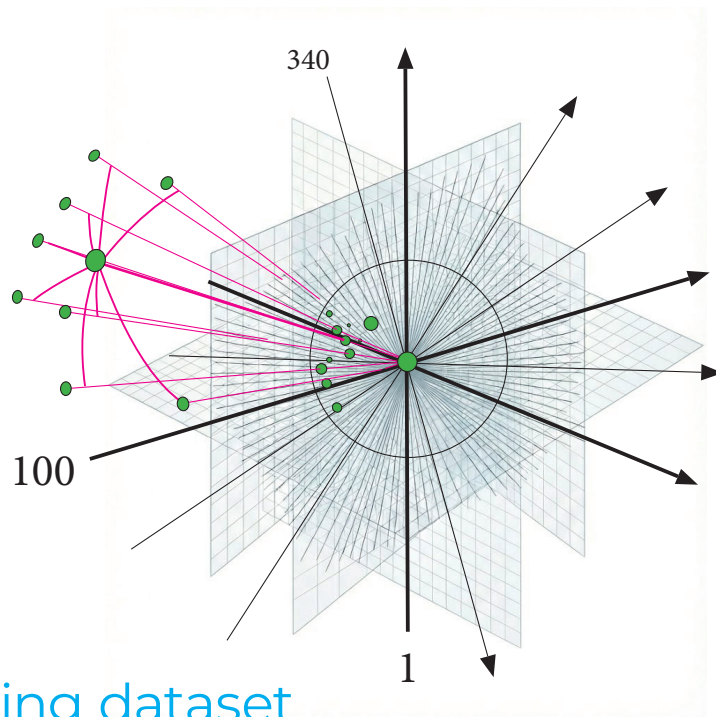


(3) Aditya, Cunha, Bunsen, Cañas, Pasi, Pinoy, Helsing, Russo, Botham, Sabourin, Fréchette, Ancil, Lopez, Navarro, Perez Pimentel, Zamora, Silva, Gagnon, August, Bjerger, Segura, Bélisle, Basset, McFarland, Roy, Høye, Larrivée, Rolnick. 2024. "Insect Identification in the Wild: The AMI Dataset."

Deep Nearest Neighbors – (91 ID vs. 37 OOD)

defining thresholds

Threshold based differentiation between ID and OOD Samples



Training dataset

knowledge storage

L2 normalization

■ over_all_examples based on validation dataset

every correct predicted Sample in Validation dataset

save distance to k-th neighbour in a list

sort distances and take value by percentile

list = [0.15, 0.17, 0.23, 0.26, 0.29, 0.32, 0.52, 0.63, 0.66, 0.73]



■ for_each_species based on validation dataset

save distance to k-th neighbour for each class in a separated list

sort distances and take value by percentile for each class

■ in_each_species based on knowledge storage

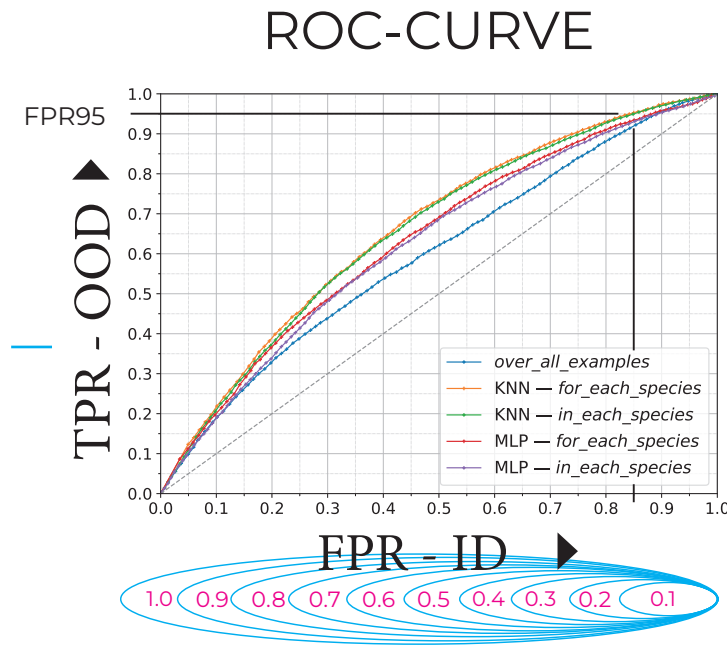
take distance to k-th neighbour in each class in a separated list

sort distances and take value by percentile for each class

Deep Nearest Neighbors – (91 ID vs. 37 OOD)

Test dataset

High Res.



	AUROC	FPR95
over_all...	0.5963 (-0.0562)	89.03% (+4.71)
combined with KNN		
for_each...	0.6523 (-0.0002)	84.32%

Deep Nearest Neighbors – (91 ID vs. 37 OOD)

Test dataset

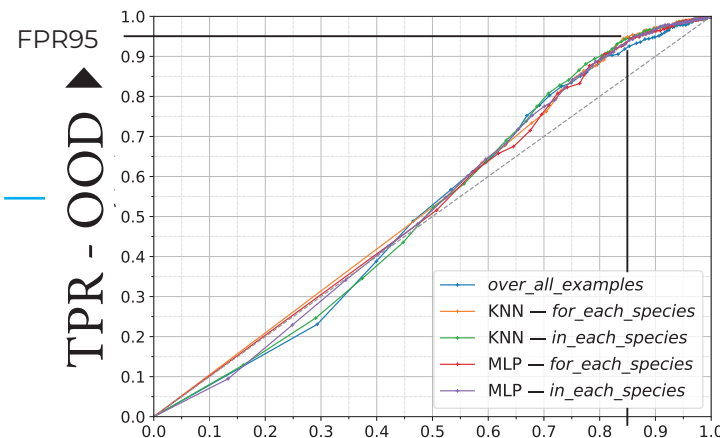
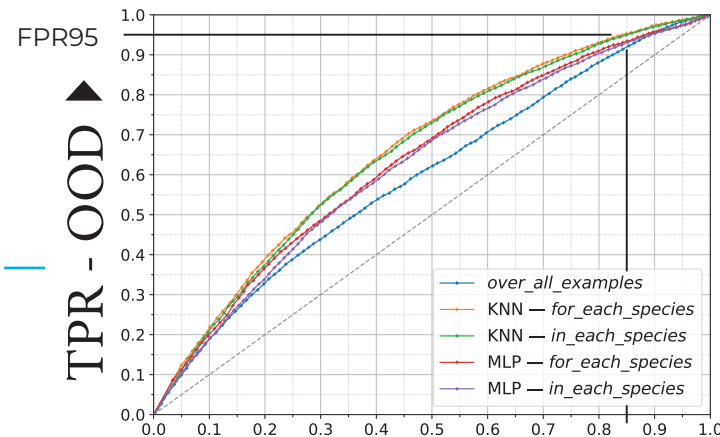
High Res.



Low Res.



ROC-CURVE



FPR - ID

AUROC

FPR95

over_all... 0.5963 (-0.0562) 89.03% (+4.71)

combined with KNN

for_each... 0.6523 (-0.0002) 84.32%

AUROC

FPR95

over_all... 0.5169 (-0.1356) 90.31% (+5.28)

combined with KNN

for_each... 0.5350 (-0.1175) 85.21% (+0.89%)

& for Fine-Grained Species Classification Domain Generalisation

MLP

Training dataset

KNN

Test dataset

Test dataset
SquaredTest dataset
Squared

Test dataset

High Res.

High Res.

High Res.

High Res.

Low Res.

Low Res.

Low Res.

Low Res.

(-2.01%)

92.32%

(-10.65%)

81.67%

(-4.54%)

(+0.39%)

43.78%

(-8.24%)

35.54%

(+2.03%)

94.33%

(-8.12%)

86.21%

43.39%

(-9.88%)

33.51%

(-50.94%)

(-52.7%)

and Novelty Detection

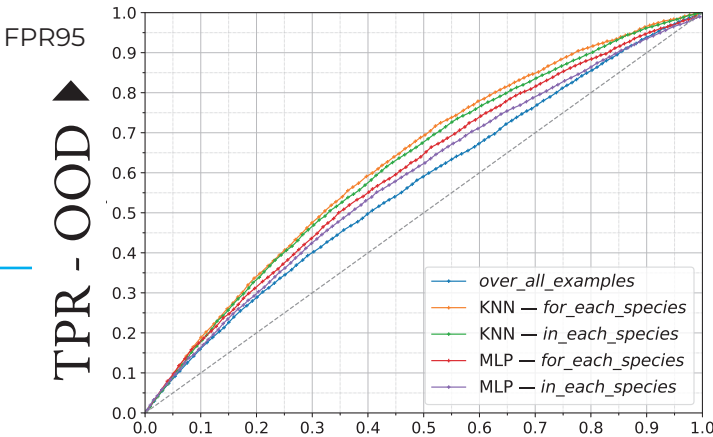
with DKNN

Test dataset

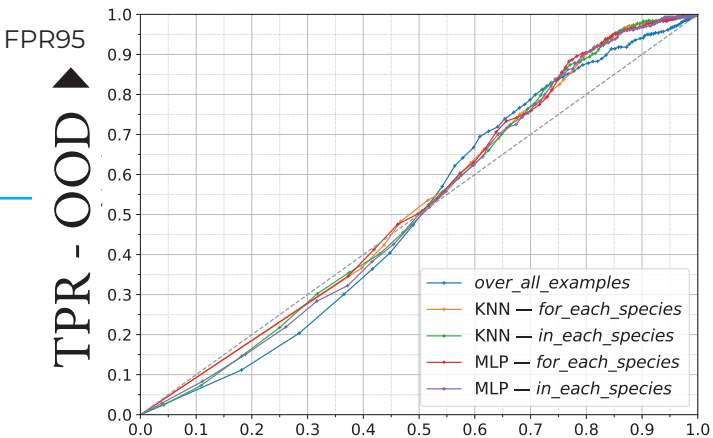
High Res.



ROC-CURVE



Low Res.



TPR - OOD

AUROC

FPR95

over_all... 0.5681 (-0,0282) 91.69% (+2.66)

combined with KNN

for_each... 0.6209 (-0.0316) 87.62% (+3.3)

in_each.. 0.6172 (-0.0353) 87.93% (+2.9)

over_all... 0.5025 (-0,0144) 91.55% (+1.24)

combined with MLP

for_each... 0.5243 (-0,0039) 85.04% (+2.22)